

CPA-Security and Pseudorandom Functions

CS 171: Cryptography

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Overview

- 1 Multiple-Message Security
- 2 CPA-Security
- 3 Pseudorandom Functions
- 4 CPA-Secure Encryption from PRFs
- 5 CPA-Security vs. Mult-Security
- 6 Summary

Pseudo-OTP: Recap

Pseudo-OTP

$\text{Enc}_k(m) = G(k) \oplus m$, where $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ is a PRG.

- Pseudo-OTP **is secure** (EAV-secure for a single message),
 - assuming G is a PRG,
 - with respect to our (indistinguishability) definition.
- **Gain**: pseudo-OTP has a *short* key — n bits instead of $\ell(n)$ bits.

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Next question

Does pseudo-OTP allow encryption of *multiple* messages? Let's first define what that means!

Security for Multiple Messages

- So far: security for a **single** encrypted message.
- In practice, Alice and Bob send **many** messages under one key.
- There are *several ways* to define security for multiple messages!

Today

- **CPA-security**: indistinguishability under a chosen-plaintext attack — the de facto minimum notion.

CPA-Security (De Facto Minimum)

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}(n)$

- 1 $k \leftarrow \text{Gen}(1^n)$. $\mathcal{A}$ is given 1^n and **oracle access** to $\text{Enc}_k(\cdot)$.
- 2 $\mathcal{A}^{\text{Enc}_k(\cdot)}$ outputs $m_0, m_1 \in \{0, 1\}^*$ with $|m_0| = |m_1|$.
- 3 $b \leftarrow \{0, 1\}$, $c^* \leftarrow \text{Enc}_k(m_b)$.
- 4 c^* is given to $\mathcal{A}^{\text{Enc}_k(\cdot)}$.
- 5 $\mathcal{A}^{\text{Enc}_k(\cdot)}$ outputs a guess b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

Definition 1 (CPA-Security)

Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has **indistinguishable encryptions under chosen-plaintext attack** (is *CPA-secure*) if for all PPT $\mathcal{A}$:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

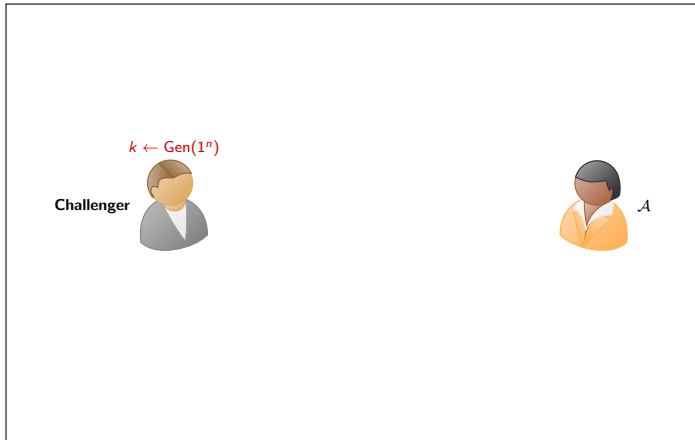
$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}$: Challenger vs. Adversary

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{CPA}}(n)$

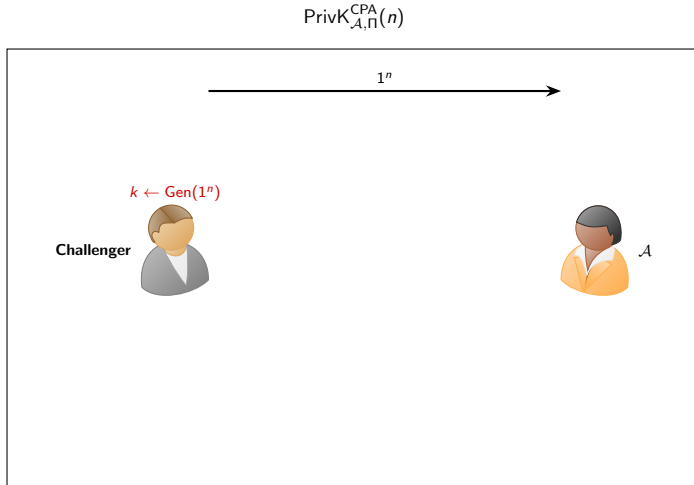


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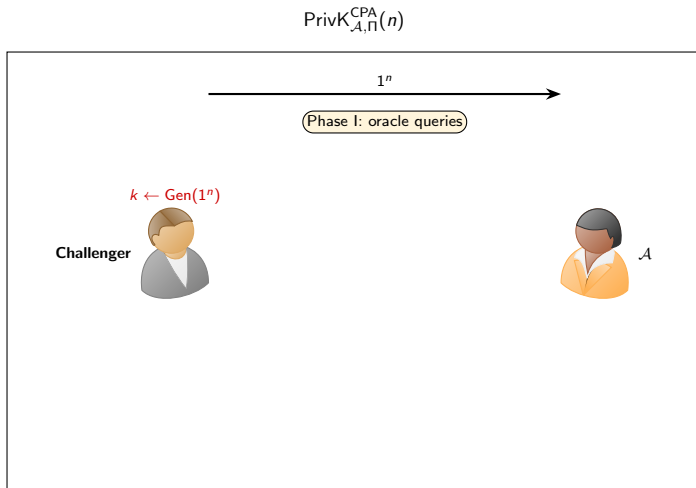
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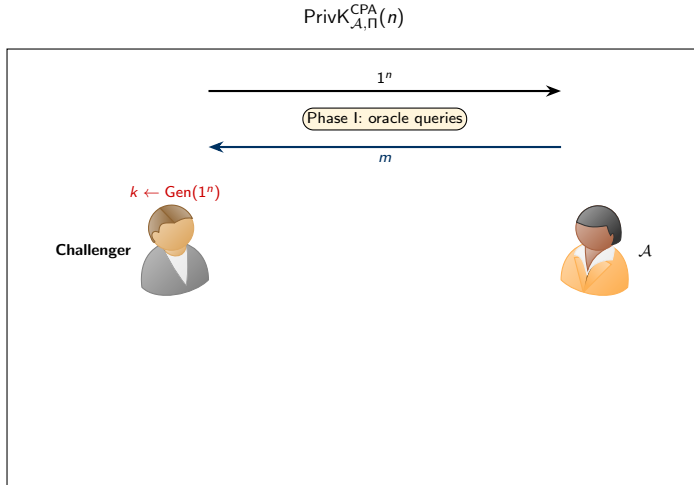
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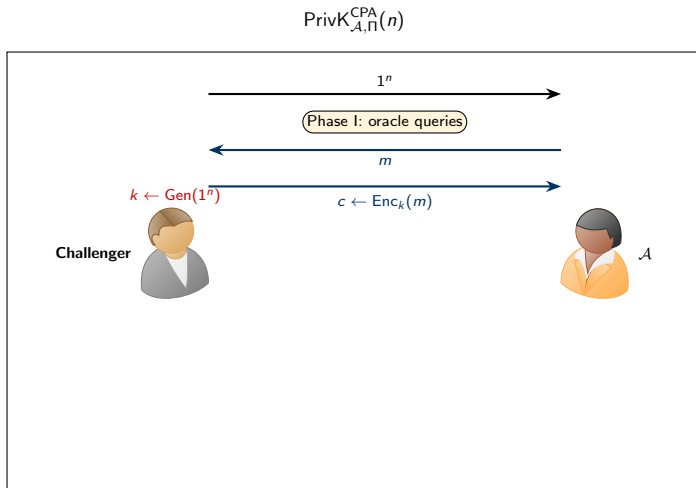
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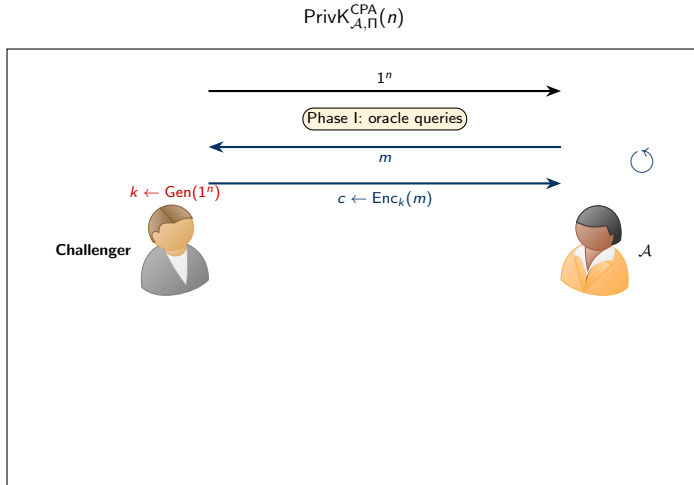
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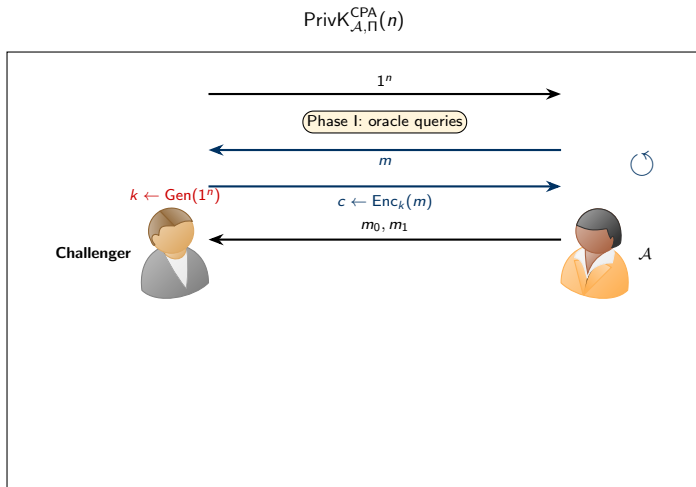
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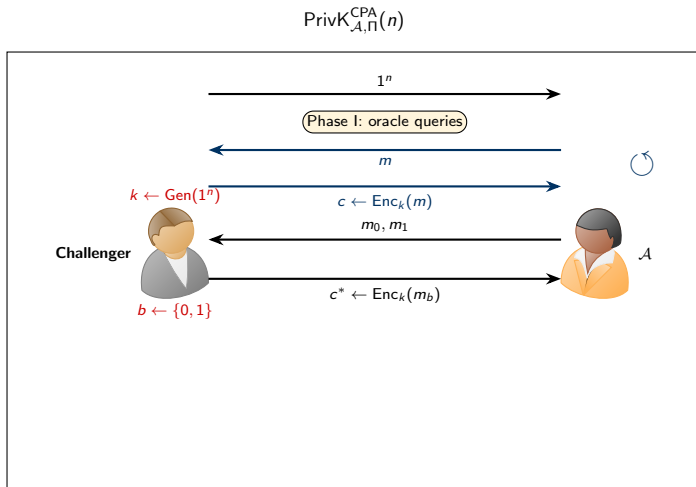
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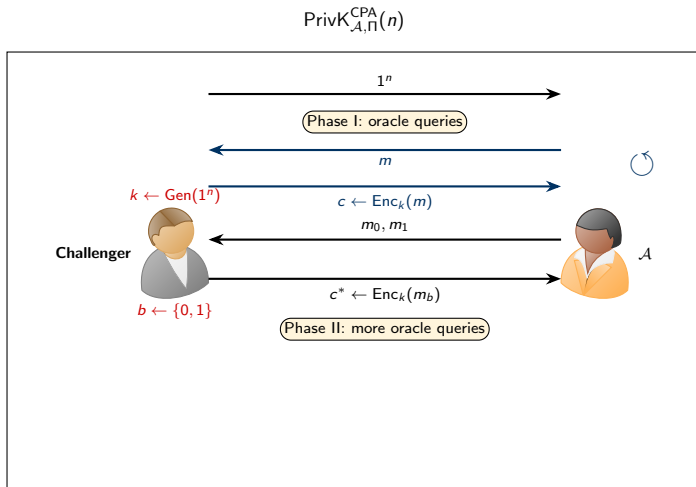
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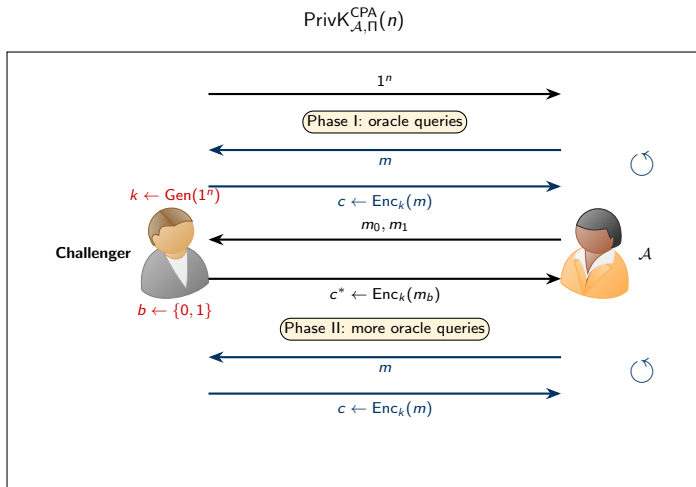
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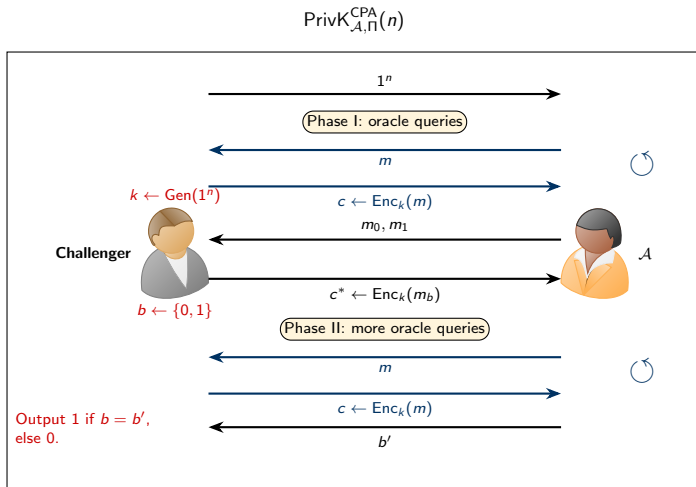
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- 2 $\mathcal{A}$ submits challenge messages $m_0 = 0^\ell$ and $m_1 = 1^\ell$.
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Theorem

Any (stateless) encryption scheme in which Enc is a *deterministic* function of the key and the message **cannot** be CPA-secure.

CPA-Security from Multiple Encryptions

- One can define other “seemingly” stronger notions of CPA-security (e.g. challenging on *vectors* of messages).
- It turns out these notions are all **equivalent** to (single-message) CPA-security.

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Consequences

- CPA-security makes it **easy to encrypt long messages**, block by block:

$$\text{Enc}_k(m_1 \parallel \cdots \parallel m_\ell) = \text{Enc}_k(m_1) \parallel \cdots \parallel \text{Enc}_k(m_\ell).$$

- **No deterministic** (stateless) encryption scheme can be CPA-secure — encryption must be *randomized*.

Detour: **Pseudorandom Functions** (a building block)

- To build a CPA-secure scheme we need encryption to be *randomized*.
- Key tool: a **pseudorandom function** (PRF) — a keyed function that behaves like a truly random function.

First, What Is a Random Function?

- Choose a **uniformly random** function (from the set of *all* functions) and then interact with this fixed function.
- Once the function has been chosen, there is **no additional randomness** involved — it is a fixed (though huge) object.

Set of All Functions Func_n

Func_n is the set of all functions from $\{0, 1\}^n$ to $\{0, 1\}^n$.

- **How many functions are there in Func_n ?**
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- So $|\text{Func}_n| = 2^{n \cdot 2^n}$.
- Sampling a random function involves sampling one of the functions in Func_n at random and fixing it.
- Sometimes useful to sample the function “**on the fly**” (lazily, answering each new query with fresh randomness).

Pseudorandom Function (PRF)

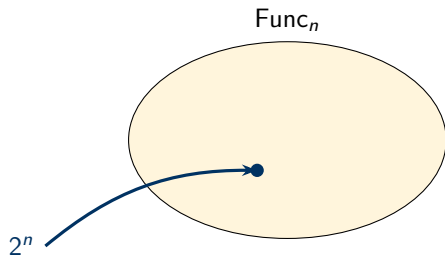
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A PRF is sampled from a space of only 2^n functions (one per key k) inside Func_n .

Keyed Functions

Definition 2 (Keyed Function)

A **keyed function** $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$, where n is the security parameter, takes two inputs:

- the first input is the **key** k ,
 - the second is the **input** x (we write $F_k(x) := F(k, x)$).
-
- Key, input, and output lengths could differ, but we will use n for simplicity.
 - F_k will be the sampled function, which we will claim is pseudorandom: on input x , the output is $F_k(x) = F(k, x)$.
 - We are only interested in **efficiently computable** $F(\cdot, \cdot)$.

Pseudorandom Function (PRF): Definition

Game $\text{PRF}_{\mathcal{A}, F}(n)$

- ① $b \leftarrow \{0, 1\}$.
- ② If $b = 0$: $k \leftarrow \{0, 1\}^n$ and set $\mathcal{O} := F_k(\cdot)$; if $b = 1$: $f \leftarrow \text{Func}_n$ and set $\mathcal{O} := f(\cdot)$.
- ③ $\mathcal{A}$ is given 1^n and **oracle access** to $\mathcal{O}(\cdot)$: it may query any $x \in \{0, 1\}^n$ and receives $\mathcal{O}(x)$.
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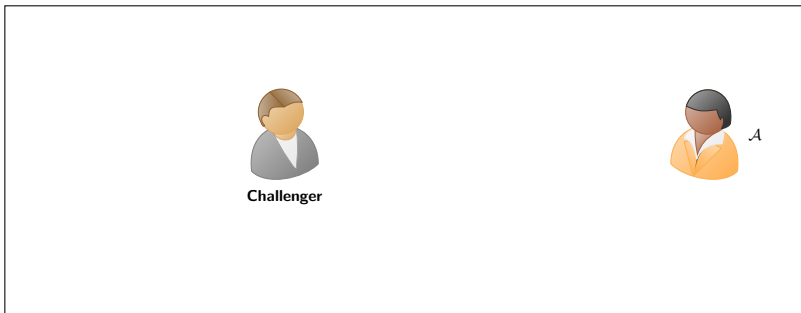
Definition 3 (Pseudorandom Function)

Let $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ be an **efficient, length-preserving, keyed** function. F is a **PRF** if for all PPT $\mathcal{A}$:

$$\Pr[\text{PRF}_{\mathcal{A},F}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

$\text{PRF}_{\mathcal{A},F}$: Challenger vs. Adversary

$\text{PRF}_{\mathcal{A},F}(n)$



PRF _{$\mathcal{A}, F$} : Challenger vs. Adversary

PRF _{$\mathcal{A}, F$} (n)

$b \leftarrow \{0, 1\}$
if $b = 0$: $\mathcal{O} := F_k, k \leftarrow \{0, 1\}^n$
if $b = 1$: $\mathcal{O} := f \leftarrow \text{Func}_n$



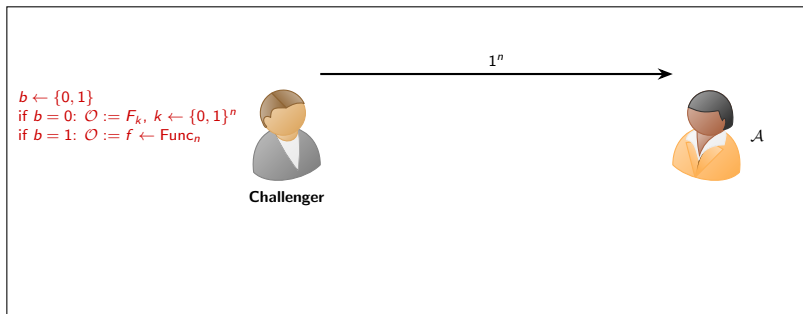
Challenger



$\mathcal{A}$

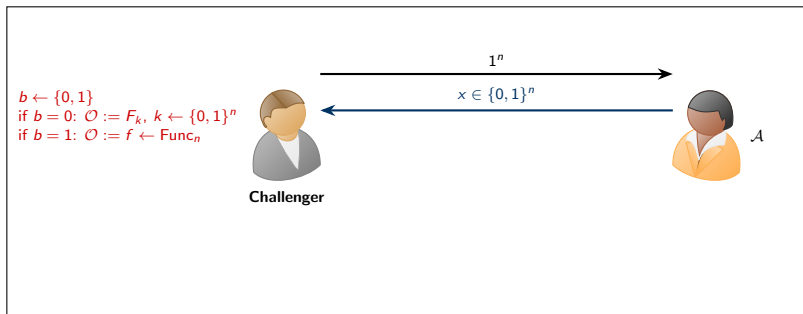
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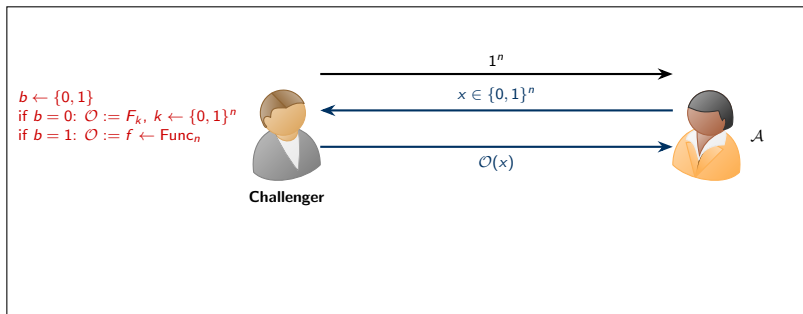
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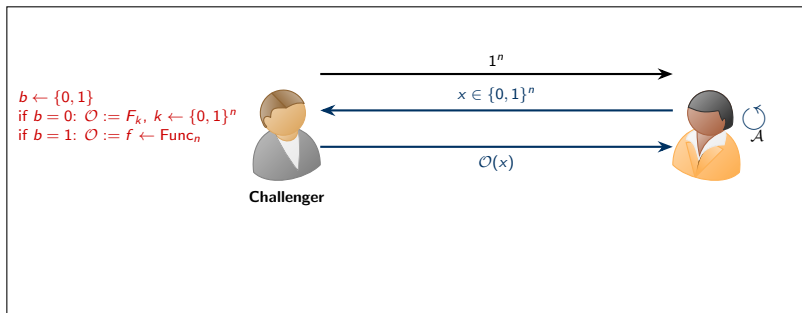
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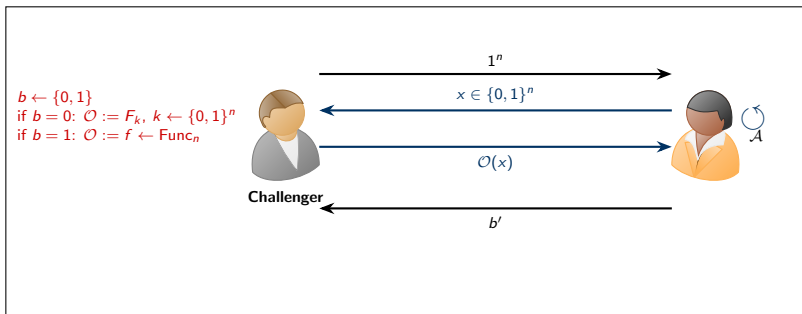
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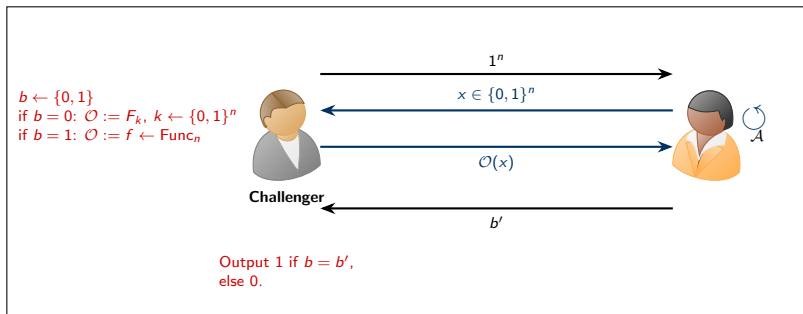
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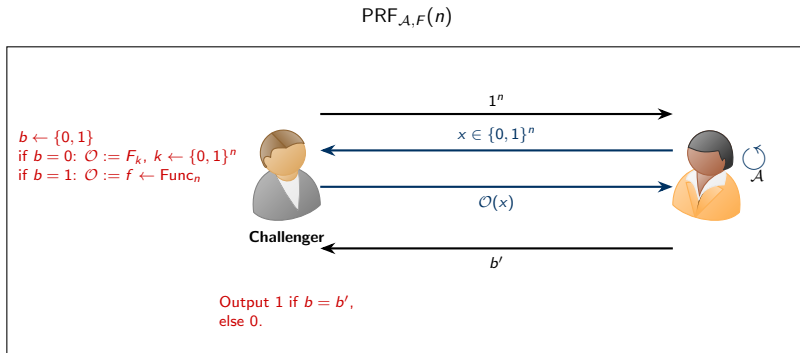


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PRF _{$\mathcal{A}, F$} : Challenger vs. Adversary



Remark

$\mathcal{A}$ interacts with either F_k (for a uniform, secret k) or a truly random f , and must guess which. For a PRF, no PPT $\mathcal{A}$ can tell the two worlds apart.

Is This a Secure PRF?

$$F(k, x) = k \oplus x?$$

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No — there is an adversary that wins the PRF game

For any two queries x_1, x_2 :

$$F(k, x_1) \oplus F(k, x_2) = (k \oplus x_1) \oplus (k \oplus x_2) = x_1 \oplus x_2.$$

- $\mathcal{A}$ queries $x_1 \neq x_2$ and checks whether the answers XOR to $x_1 \oplus x_2$ — true with probability 1 for this candidate.
- This would **not** be the case for a random function f : there $f(x_1) \oplus f(x_2) = x_1 \oplus x_2$ only with probability 2^{-n} .

Do PRFs Exist?

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- **But:** we know how to *construct* PRFs from PRGs.

Takeaway

PRFs exist if PRGs exist.

CPA-Secure Encryption from a PRF

Construction

Let F be a PRF, $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$. Define $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$:

- $\text{Gen}(1^n)$: choose uniform $k \leftarrow \{0, 1\}^n$ and output it as the key.
- $\text{Enc}_k(m)$ (for $m \in \{0, 1\}^n$): sample uniform $r \leftarrow \{0, 1\}^n$ and output the ciphertext

$$c := \langle r, F_k(r) \oplus m \rangle.$$

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Key point

The encryption scheme is **randomized**! (Fresh r per message — exactly what CPA-security requires.)

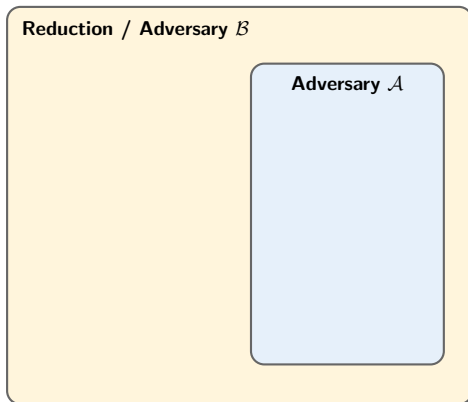
Theorem 4

*If F is a PRF, then the construction on the previous slide is a **CPA-secure** encryption scheme.*

Proof by reduction

Given a PPT adversary $\mathcal{A}$ “breaking” CPA-security of the encryption scheme, construct a PPT adversary $\mathcal{B}$ “breaking” the PRF F .

Proof of Security: The Reduction $\mathcal{B}$

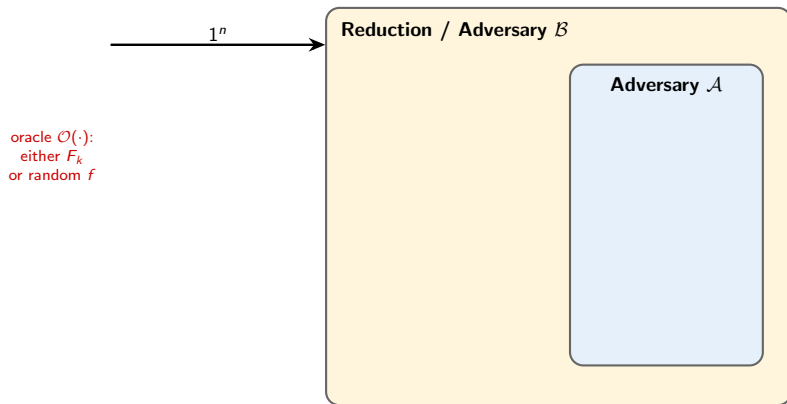


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To prove: $\Pr[\text{PRF}_{\mathcal{B}, F}(n) = 1] \geq \frac{1}{2} + \delta(n)$ for non-negligible δ .

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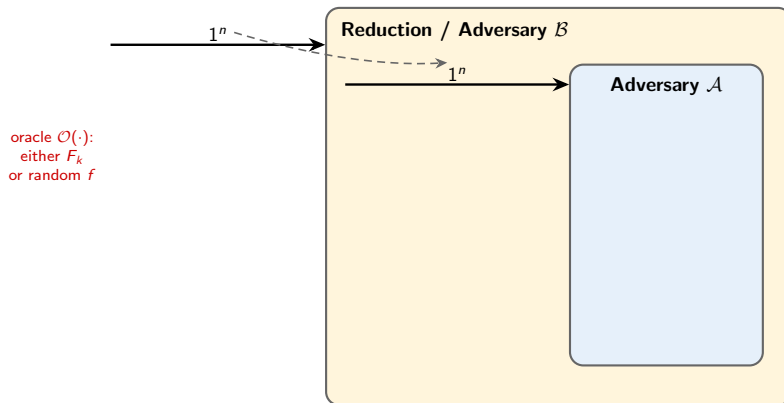


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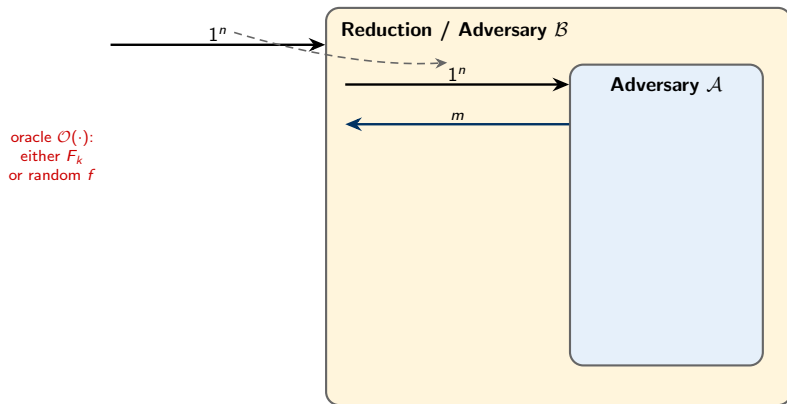


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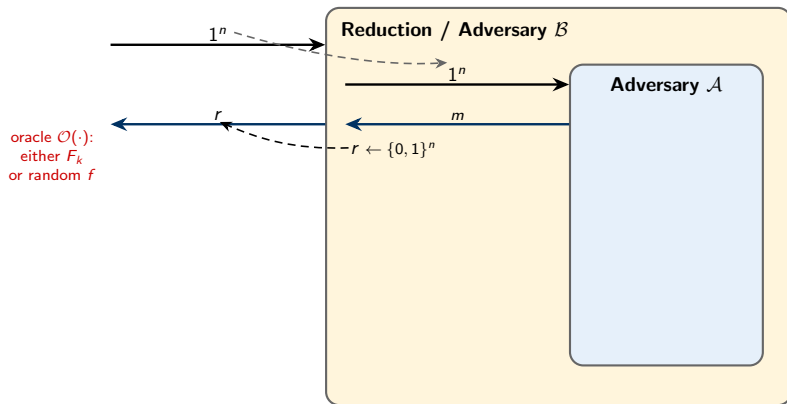


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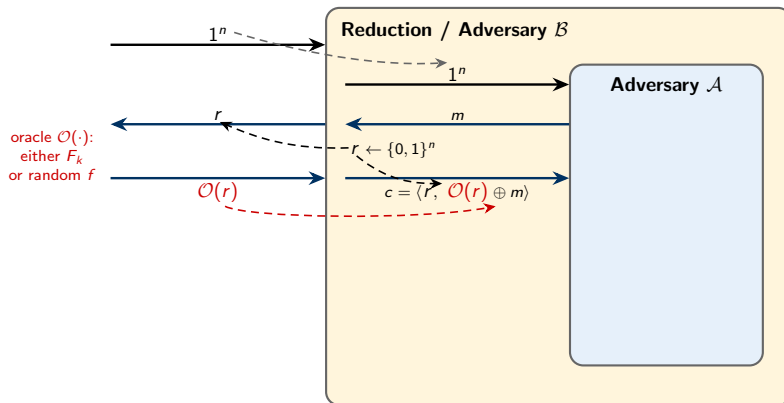


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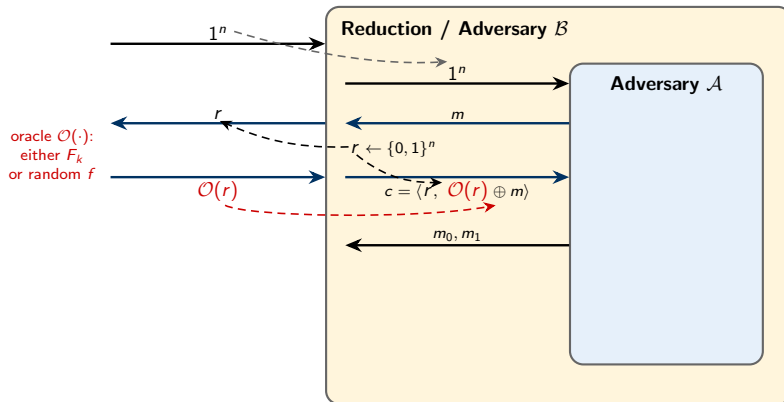


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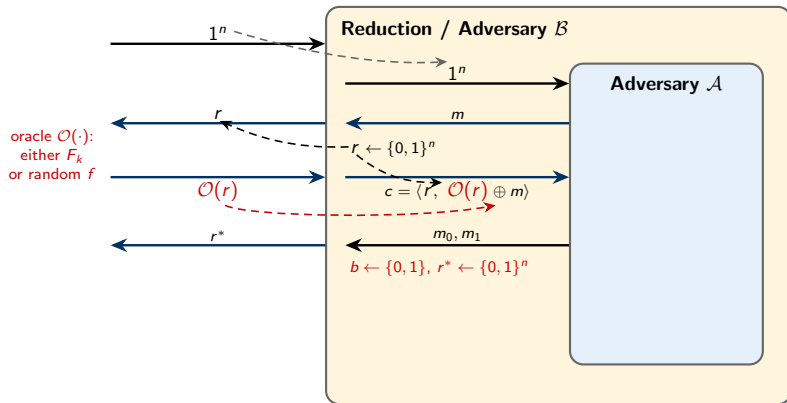


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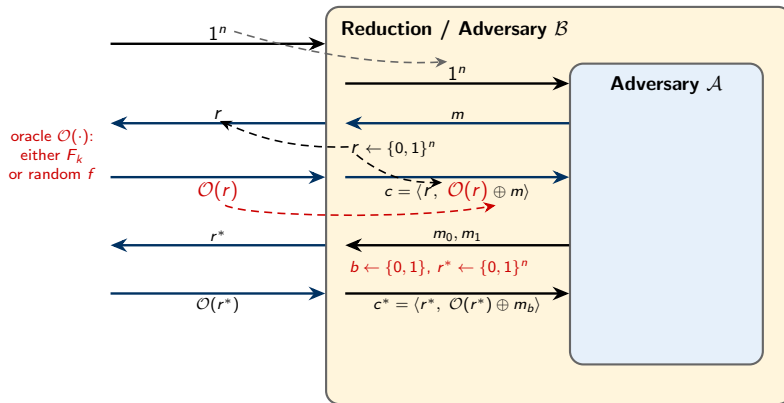


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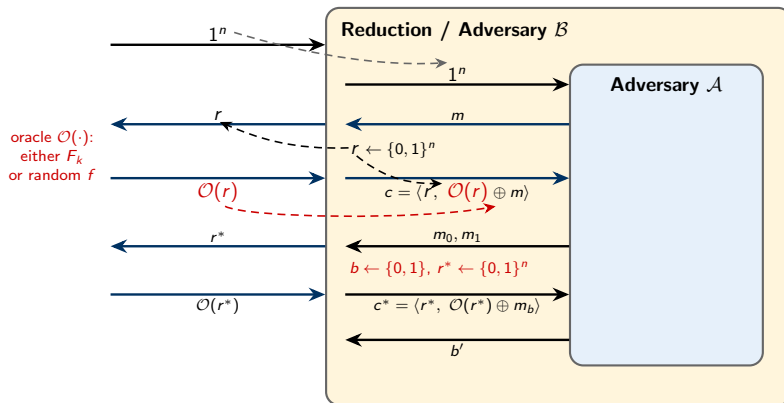


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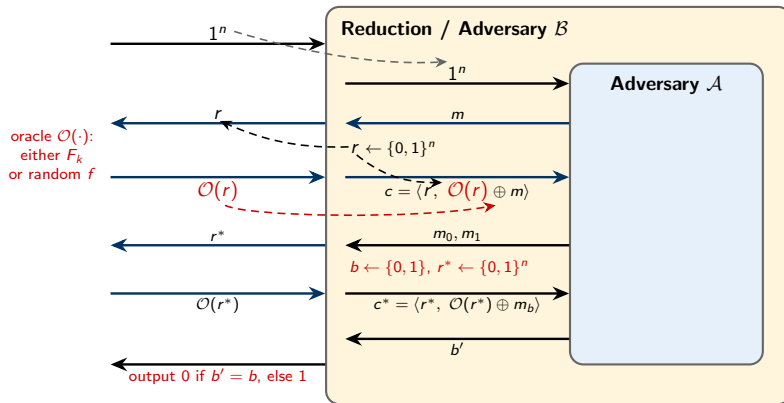


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Proof of Security: The Reduction $\mathcal{B}$ (in Words)

Recall $\text{Enc}_k(m)$: sample $r \leftarrow \{0, 1\}^n$ and output $c := \langle r, F_k(r) \oplus m \rangle$.

$\mathcal{B}$, given 1^n and oracle access to $\mathcal{O}(\cdot)$

$\mathcal{B}$ runs $\mathcal{A}(1^n)$ and plays the CPA challenger for it, using $\mathcal{O}$ wherever the challenger would use F_k :

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- $\mathcal{B}$ never needs k — oracle access to $\mathcal{O}$ suffices to simulate the challenger.
- $\mathcal{B}$ is PPT, given $\mathcal{A}$ is PPT.

Proof of Security: Two Cases

Case $\mathcal{O} = F_k$ (pseudorandom)

$\mathcal{A}$'s view is the real scheme, so
 $\Pr[b = b'] \geq \frac{1}{2} + \epsilon(n).$

Case $\mathcal{O} = f$ (truly random)

Like a one-time pad, unless r^* repeats:
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Let $q(n)$ bound the number of encryption queries. If $r^* \neq r_i$ for all query values r_i , then $f(r^*)$ is uniform and independent of $\mathcal{A}$'s view, so c^* is a one-time pad encryption of m_b and $\Pr[b = b'] = \frac{1}{2}$. A collision $r^* = r_i$ happens with probability at most $q(n)/2^n$.

Proof of Security: Putting It Together

Success probability of $\mathcal{B}$

In the random case $\mathcal{B}$ is correct exactly when $\mathcal{A}$ loses:

$$\Pr[\mathcal{B} \text{ correct} \mid \mathcal{O} = f] = \Pr[b' \neq b \mid \mathcal{O} = f] = 1 - \Pr[b' = b \mid \mathcal{O} = f] \geq \frac{1}{2} - \frac{q(n)}{2^n}.$$

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If $\epsilon(n)$ is non-negligible then so is $\epsilon(n)/2 - q(n)/2^{n+1}$ (since q is polynomial), contradicting that F is a PRF. □

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Remark (where randomness helped)

The fresh r per encryption is what makes $f(r^*)$ “unused” with overwhelming probability — this is exactly where the deterministic pseudo-OTP failed.

- **Gets CPA security** (proved above).
- Can encrypt messages of **arbitrary length**, block by block:

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Negative: $\text{Enc}_k(m) = \langle r, F_k(r) \oplus m \rangle$

Ciphertext size is double the message length (we send r alongside $F_k(r) \oplus m$).

Multi-Security: A Weaker Multi-Message Notion

Game $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$

- 1 $\mathcal{A}$ is given the security parameter 1^n .
- 2 For each $i \in \{1, \dots, t\}$, $\mathcal{A}$ outputs $m_{0,i}, m_{1,i} \in \{0, 1\}^*$ with $|m_{0,i}| = |m_{1,i}|$.
- 3 $b \leftarrow \{0, 1\}$, $k \leftarrow \text{Gen}(1^n)$, $c_i \leftarrow \text{Enc}_k(m_{b,i})$.
- 4 $c_1, \dots, c_t$ is given to $\mathcal{A}$.
- 5 $\mathcal{A}$ outputs a guess b' .
- 6 Output 1 if $b = b'$, and 0 otherwise.

Definition 5 (Multi-Security)

Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has **indistinguishable multiple encryptions** in the presence of an eavesdropper (is *mult-secure*) if for all PPT $\mathcal{A}$:

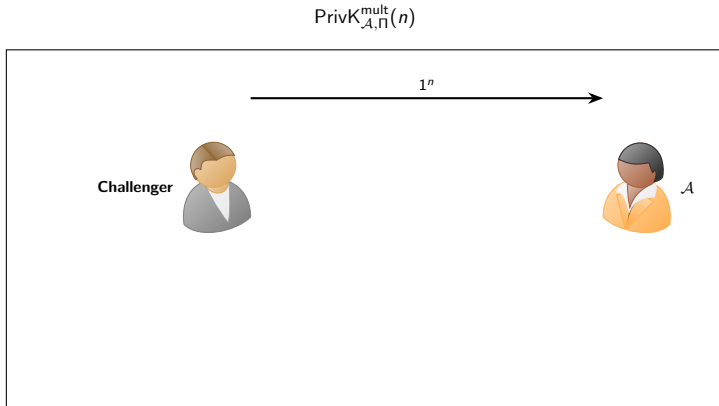
$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}$: Challenger vs. Adversary

$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}(n)$

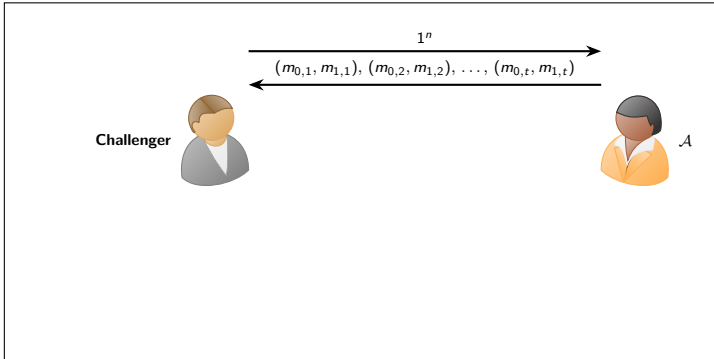


$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}$: Challenger vs. Adversary



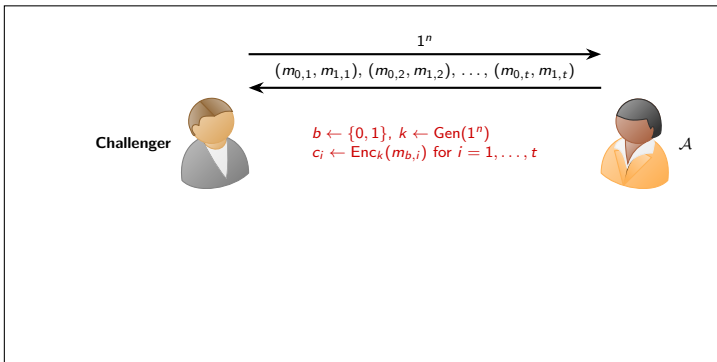
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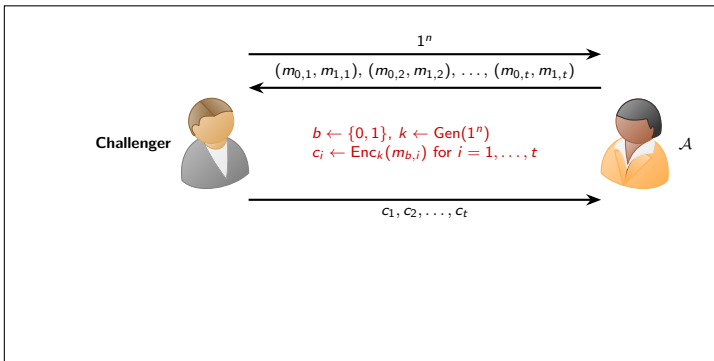
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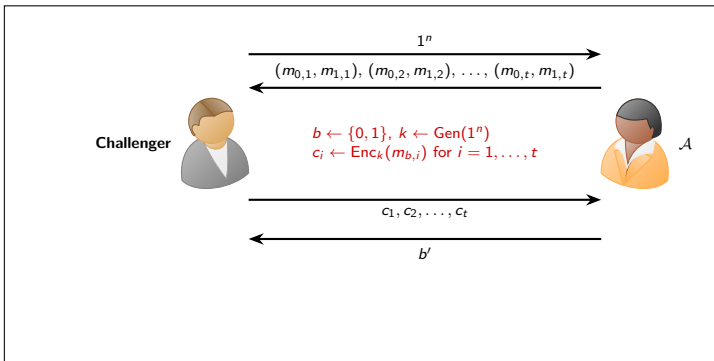
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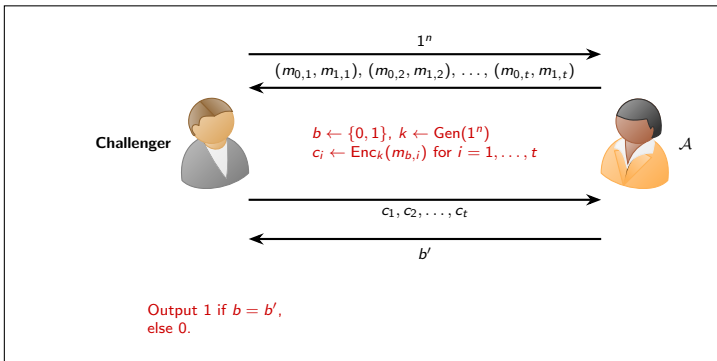
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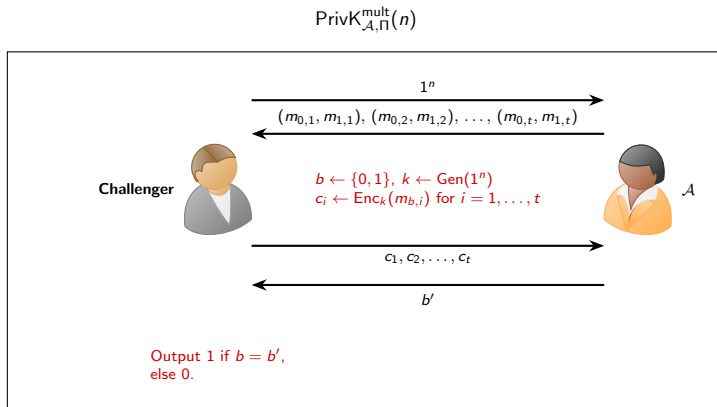


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$\text{PrivK}_{\mathcal{A}, \Pi}^{\text{mult}}$: Challenger vs. Adversary



Remark

Same as $\text{PrivK}^{\text{eav}}$, except $\mathcal{A}$ submits t message pairs at once (all under the *same* key k). The pairs are chosen **up front**: $\mathcal{A}$ sees no ciphertext before committing to them.

CPA-Security Is Stronger Than Mult-Security

- **How can we prove this?**

- Construct an encryption scheme Π that is **mult-secure** but **not CPA-secure**.
- *Simplify the problem:* assume Φ is mult-secure *and* CPA-secure; we weaken Φ to get Π so that Π is mult-secure but not CPA-secure.

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The (weird) scheme Π

Given $\Phi = (\text{Gen}, \text{Enc}, \text{Dec})$, set $\Pi = (\text{Gen}', \text{Enc}', \text{Dec}')$:

- $\text{Gen}'(1^n)$: set $k' := (k, m^*)$, where $k, m^* \leftarrow \text{Gen}(1^n)$.
- $\text{Enc}'_{k'}(m)$: if $m = m^*$ then output m^* . Otherwise, output $\text{Enc}_k(m) \parallel m^*$.
- $\text{Dec}'_{k'}(c)$: define naturally!

Need to prove that (1) Π is **mult-secure**, but (2) Π is **not CPA-secure**! *Why are we looking at this weird scheme?*

(1) Π Is Mult-Secure

Argument

- In the *mult* (eavesdropper) game, $\mathcal{A}$ has **no encryption oracle** — it only submits its message pairs up front.
- The probability that $\mathcal{A}$ asks for an encryption of m^* is **negligible** (at most $\frac{2t}{2^n}$), since the secret key has at least n bits and $\mathcal{A}$ does not know m^* .
- If there are no such “weird” queries, the game is the **same** as the mult-game for Φ , which is mult-secure.

Hence Π is mult-secure. □

(2) Π Is Not CPA-Secure

CPA attack $\mathcal{A}$

- 1 Query $\text{Enc}'_{k'}(\cdot)$ on input 0^n , and let $c \parallel m^*$ be the received ciphertext.
- 2 Submit challenge messages $m_0 = m^*$ and $m_1 = 0^n$.
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 - ② Submit challenge messages $m_0 = m^*$ and $m_1 = 0^n$.
 - ③ Output 0 if $c^* = m^*$, and 1 otherwise.
- If $b = 0$: $c^* = \text{Enc}'_{k'}(m^*) = m^*$, so $\mathcal{A}$ outputs 0.
 - If $b = 1$: $c^* = \text{Enc}_k(0^n) \parallel m^* \neq m^*$, so $\mathcal{A}$ outputs 1.
 - $\mathcal{A}$ wins with probability 1 (whenever $m^* \neq 0^n$) — the encryption oracle is available in the CPA game but *not* in the mult game.

Summary

- **Multiple-message security:** CPA-security (chosen-plaintext, oracle access to Enc_k) is the de facto minimum.
- **Pseudo-OTP is not CPA-secure** — in fact no deterministic stateless scheme is; CPA needs **randomized** encryption.
- **Pseudorandom function (PRF):** a keyed function F_k indistinguishable from a truly random function $f \leftarrow \text{Func}_n$.
- **CPA-secure encryption from a PRF:** $\text{Enc}_k(m) = \langle r, F_k(r) \oplus m \rangle$; secure by reducing to PRF security (two cases).
 - Cost: ciphertext is twice the message length.
- **CPA-security is strictly stronger than mult-security** (the eavesdropper notion for t messages chosen up front): a mult-secure scheme Φ can be weakened into Π that is mult-secure but not CPA-secure.